



# What does a medicine's price **pay for?**

Part 1 showed the active ingredient became cheaper over four decades. This part reviews the rest of the cost structure — **eight companies' SEC filings** (SEC: the US Securities and Exchange Commission, the regulator every listed company reports to), human pharma and animal health side by side. The difference explains a twenty-year price divergence.

# One industry code, **two cost structures.**

R&D • SHARE OF REVENUE

**19% vs 7%**

Human pharma median **19.4%**; animal health **7.3%** — a **2.7x** difference in the share of revenue reinvested in R&D.

COST OF GOODS • SHARE OF REVENUE

**30% vs 45%**

The ranking inverts: human **29.6%**, animal **44.6%**. Animal health spends the larger share on manufacturing, where the ingredient sits.

WHAT REMAINS AFTER COGS+R&D+SG&A

**~30 vs ~23**

Per \$100 of revenue: human ~30, animal ~23. **Similar remainders, reached through different structures.**

**This is consistent with the price record:** an R&D-weighted cost structure finances future products through price — human prices rose +75% real over twenty years. A manufacturing-weighted structure follows its input costs — veterinary prices fell 4% real over the same window. **The cost structure and the price behaviour agree.**

# Straight from the filings. Nothing modelled.



## Source

Annual 10-K figures via **SEC EDGAR's XBRL API** (SEC: US Securities and Exchange Commission; the 10-K is the audited annual report every US-listed company must file) — machine-readable, public, keyless.



## Measure

COGS, R&D and SG&A as % of revenue — **mean of FY2023-25** per company, **median** across each group so no single filer drags the figure.



## Two clean groups

**Human:** Pfizer, Lilly, AbbVie, Bristol Myers Squibb. **Animal:** Zoetis, Elanco, Phibro — pure-play filers only. Merck (~9% animal health) is shown but **pooled with neither group**. Animal-health bars are shown **unattributed**: the finding is the group contrast, not any one filer's margin. All figures re-derivable from the filings above.

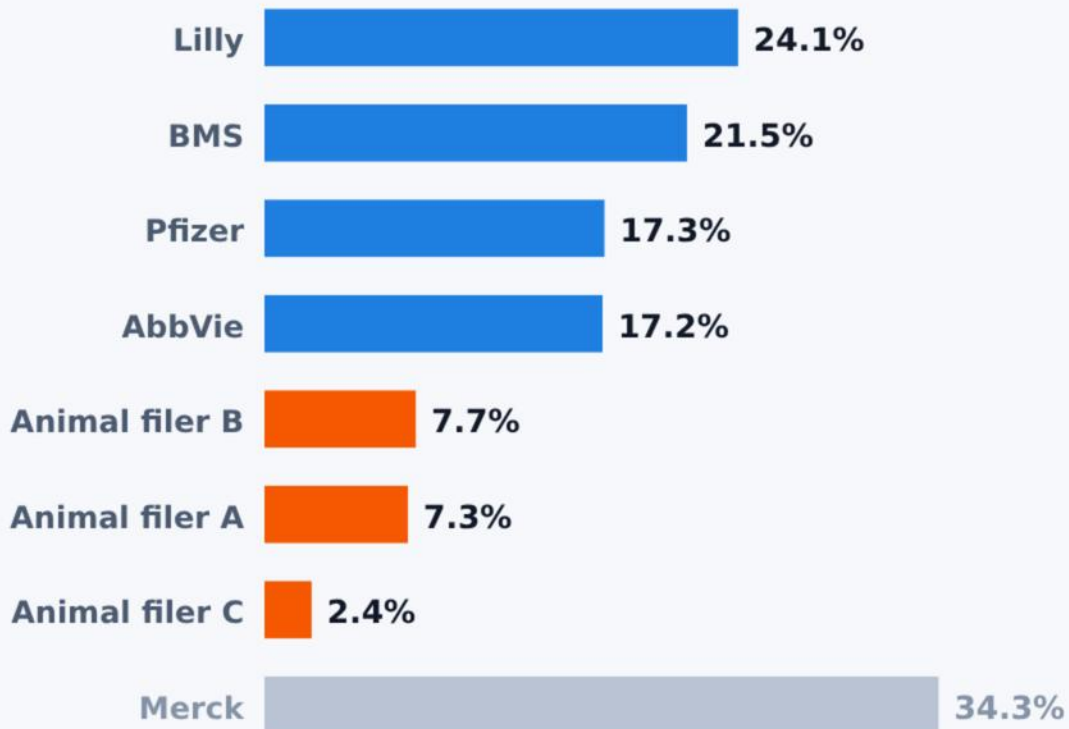


## What COGS is not

**COGS is not the ingredient.** No filer splits API out of its cost of goods — the ingredient from Part 1 sits *inside* these bars, unquantified.

# R&D share: the main difference.

R&D as a share of revenue, FY2023–25 average. Every human filer sits between 17% and 24%. Every animal filer sits **below 8%**.

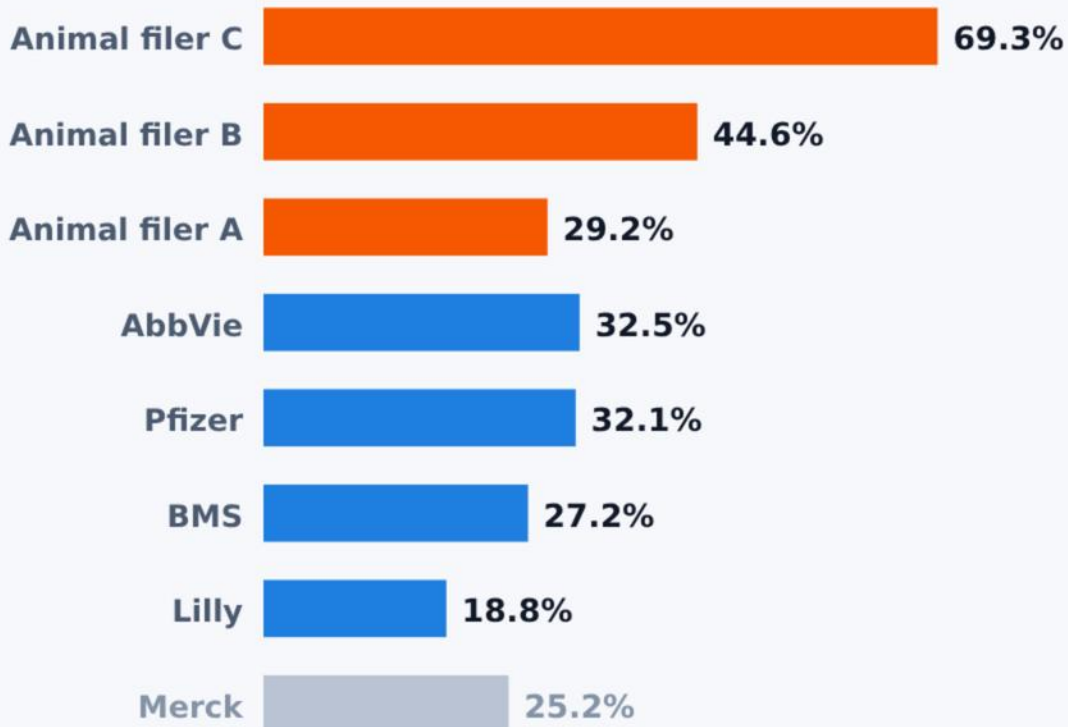


SEC 10-K FILINGS, FY2023–25 AVG • MERCK'S 34.3% INCLUDES LARGE ACQUIRED-IPR&D CHARGES – ONE REASON IT POOLS WITH NEITHER GROUP



# Cost of goods: the ranking inverts.

Cost of goods as a share of revenue. Animal health runs **29–69%**; human pharma 19–33%. **A high COGS share does not mean costs rose — it means price sits close to cost. This is consistent with four decades of veterinary prices tracking inflation.**

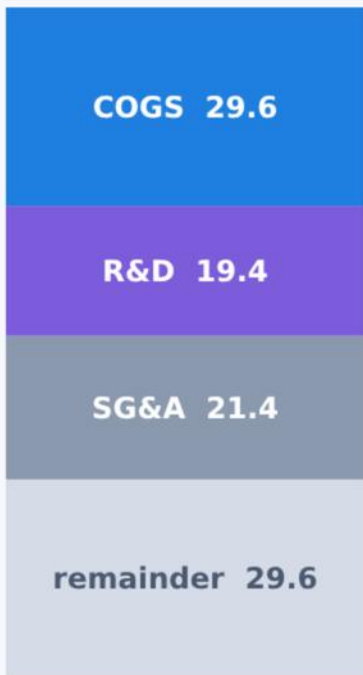


SEC 10-K FILINGS, FY2023–25 AVG • SPREAD INSIDE ANIMAL HEALTH IS REAL: 29% TO 69% ACROSS THE THREE FILERS — PETCARE AND NUTRITION ARE DIFFERENT BUSINESSES

# Cost structure per \$100 of revenue.

Group medians, side by side. **Different structures** — and a similar remainder at the end of the line.

HUMAN PHARMA • MEDIAN OF 4



ANIMAL HEALTH • MEDIAN OF 3



MEDIANS: HUMAN N=4, ANIMAL N=3 • "REMAINDER" = 100 - COGS - R&D - SG&A: OPERATING CUSHION BEFORE OTHER LINES, NOT PROFIT

# Two cost structures, **two price histories.**

R&D-BUILT P&L • HUMAN

## +75%

Real change in the US **producer (factory-gate) price** of human-dominated pharma over the clean 20-year window (2001–2021).

**R&D is financed through price** — current revenue funds future products.

MANUFACTURING-BUILT P&L • VETERINARY

## −4%

Real change in the US **veterinary producer price** on the same window (BLS WPU0634).

**Manufacturing follows its input costs** — and those inputs became cheaper. Factory-gate, not what clinics or owners pay.

**For anyone applying human-pharma benchmarks to animal health:** the benchmark does not transfer. Veterinary prices have followed a manufacturing cost base for forty years. Both figures come from public data and both hold after inflation adjustment.

# Scope and limitations.

## Small samples, stated

Four human filers, three animal. Medians, not means; three fiscal years averaged. This is an anatomy sketch, not a census.

## Global, not US-only

10-K figures are worldwide consolidated. Nothing here is a US-market cost share.

## Heterogeneity is real

Inside animal health, COGS runs 29% to 69% across the three filers — petcare and nutrition are different businesses sharing a label.

## Accounting seams

Filers switched revenue/R&D tags mid-history (unioned, logged); acquired-IPR&D inflates some R&D lines (Merck). No headcount tag exists in XBRL — employee costs stay unmeasured.



# Same ingredient market, different businesses.

## IF YOU BENCHMARK

**Benchmark within the segment** — animal-health R&D or COGS ratios judged against human-pharma norms mislead, and the reverse.

## IF YOU BUY

**The higher the COGS share, the more input-cost management matters** — procurement timing and FX are margin levers.

## IF YOU SELL

**Veterinary prices have not exceeded inflation for four decades** — a record worth stating in customer conversations.

## Every source

**Filings** · SEC EDGAR XBRL company facts ([data.sec.gov](https://data.sec.gov)), 10-K FY values, FY2023–25: Pfizer, Eli Lilly, AbbVie, Bristol Myers Squibb · Zoetis, Elanco, Phibro · Merck (shown, not pooled). Concept tags per company recorded in the repository.

**Price histories** · BLS PPIs via FRED — veterinary preparations WPU0634, pharmaceutical preparation industry PCU325412325412, CPI-U deflated, continuous-segment analysis.

Part 1: the input ledger — 44 years of the active ingredient. Every figure in both posts is recomputed from data committed in the repository, and each carries its named source.